

Code-Layout, Readability and Reusability

Date	29 September 2026
Team ID	6ab3e6756603573ccfb3b891
Project Name	FitBuddy - AI Fitness Plan Generator Using Gemini Models
Maximum Marks	5 Marks

Code Layout Checklist

S.No	Code Quality Parameter	Followed	Remarks
1	Consistent Indentation	Yes	Python modules use standard indentation and simple function bodies.
2	Proper File Structure	Yes	app, app/ai, templates, static and tests are separated logically.
3	Meaningful Variable Names	Yes	Names such as original_plan, nutrition_tip, user_id and updated_plan describe intent.
4	Function / Method Names	Yes	generate_workout_gemini, save_user, update_plan and get_all_users_with_plans are descriptive.
5	Code Comments	Partial	Core files are readable; more inline comments could be added for complex database/AI decisions.
6	Modular Design	Yes	Routes, schemas, database and three AI generators are separated.
7	No Redundant Code	Partial	The current version is compact; further refactoring can remove small inconsistencies such as helper duplication.
8	Error Handling	Yes	Routes wrap generation and feedback operations and render an error page on failure.

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Gemini client	Python / google-genai	Shared by workout, nutrition and plan-update generators	High
2	UserInput schema	Python / Pydantic	Used by workout and nutrition generation routes	High
3	Database helpers	Python / SQLAlchemy	Shared across user and plan operations	High
4	Jinja base template	HTML / Jinja2	Provides common navigation and styling for pages	High
5	Static stylesheet	CSS	Shared across all rendered pages	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Clear app / templates / static / tests separation.
Readability	4	Small modules with descriptive names.
Reusability	4	AI client and database helpers are reused.
Documentation / Comments	3	README and documentation are strong; inline comments can grow.
Overall Score	4	Suitable for the current educational project and future modular extension.